Type MS · The difference of the roots

Answer 1 of 12 · Question 11 part b.ii · 2 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part b.ii in this mark scheme extract

11 (a)(i)	$\left(\alpha-\beta=2\sqrt{6}\Rightarrow(\alpha-\beta)^2=24\right)\Rightarrow(\alpha+\beta)^2-4\alpha\beta=24$ oe **(1)**	M1
	$\alpha^2+\beta^2=30\Rightarrow(\alpha+\beta)^2-2\alpha\beta=30$ oe **(2)**	M1
	(2) – (1) $6=2\alpha\beta\Rightarrow\alpha\beta=3$ cso	dM1A1* [4]
ALT1	$\left((\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta\Rightarrow\right)2\alpha\beta=\alpha^2+\beta^2-(\alpha-\beta)^2$ oe	M1
	$30-\left(2\sqrt{6}\right)^2=6$	M1
	$2\alpha\beta=6\Rightarrow\alpha\beta=3$	dM1A1 [4]
(ii)	$30=(\alpha+\beta)^2-2\times3\Rightarrow(\alpha+\beta)^2=36\Rightarrow\alpha+\beta=6$ $[\alpha>\beta>0]$ cso	M1A1* [2]
ALT2 (i)	$\alpha-\beta=2\sqrt{6}\rightarrow\alpha=2\sqrt{6}+\beta$	
	$\alpha^2+\beta^2=30\rightarrow\left(2\sqrt{6}+\beta\right)^2+\beta^2=30$	M1
	$2\beta^2+4\sqrt{6}\beta-6=0$ oe eg $\beta^2+2\sqrt{6}\beta-3=0$	M1
	$(\beta=)\dfrac{-2\sqrt{6}\pm\sqrt{\left(2\sqrt{6}\right)^2-4(1)(-3)}}{2}$ oe $\rightarrow\beta\left(=3-\sqrt{6}\right)$	dM1
	$\alpha=3+\sqrt{6}$	
	$\alpha\beta=\left(3+\sqrt{6}\right)\left(3-\sqrt{6}\right)=9+6\sqrt{3}-6\sqrt{3}-6=3$	A1* [4]
(ii)	$\alpha+\beta=3+\sqrt{6}+3-\sqrt{6}=6$	M1A1 [2]
(b)(i)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2(\alpha\beta)^2=30^2-2\times3^2=882$	M1A1 [2]
(b) ii)	$\alpha^4-\beta^4=\left(\alpha^2+\beta^2\right)\left(\alpha^2-\beta^2\right)=\left(\alpha^2+\beta^2\right)(\alpha-\beta)(\alpha+\beta)$	M1
	$\alpha^4-\beta^4=30\times6\times2\sqrt{6}=360\sqrt{6}$	A1 [2]
(c)	$\left(\alpha^4+\beta^4\right)+\left(\alpha^4-\beta^4\right)=2\alpha^4=882+360\sqrt{6}\Rightarrow\alpha^4=441+180\sqrt{6}$	M1A1 [2]
ALT1	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\alpha=3+\sqrt{6}$	
	$\alpha^4=\left(3+\sqrt{6}\right)^4$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
ALT2	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\beta=3-\sqrt{6}$	
	$\alpha^4=\beta^4+360\sqrt{6}=\left(3-\sqrt{6}\right)^4+360\sqrt{6}$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
		Total 12 marks

Type MS · The difference of the roots

Answer 1 of 12 · Question 11 part b.ii · 2 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part b.ii in this mark scheme extract

Part	Mark	Notes
(a) (i)	M1	For forming an equation of the form $(\alpha+\beta)^2 \pm p\alpha\beta = 24$ oe or $(\alpha+\beta)^2 \pm t\alpha\beta = 30$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correctly solving their simultaneous equations in $\alpha+\beta$ and $\alpha\beta$ to find a value for $\alpha\beta$, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
ALT1	M1	For forming an equation of the form $q\alpha\beta = \alpha^2+\beta^2-(\alpha-\beta)^2$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correct substitution into their equation, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
(a) (ii)	M1	Correctly uses either of their equations (must be of the required form) to substitute the given value of $\alpha\beta$ and obtains a value for $\alpha+\beta$
	A1*cso	For the correct value of $\alpha+\beta$
ALT2 (i)	M1	For an attempt to eliminate α or β and arrive at an unsimplified quadratic equation in one variable. Allow one error
	M1	For the correct quadratic equation
	dM1	For a **fully correct method** to solve their quadratic equation in α or β, dependent on the 1st method mark
	A1*cso	For correctly finding α and β and showing the minimum steps shown to find $\alpha\beta$
	M1	For finding $\alpha+\beta$ with their values
	A1*cso	For finding $\alpha+\beta$ with the correct values, minimum steps as shown.
(b) (i)	M1	For the correct algebra $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2(\alpha\beta)^2$, if they use more complex algebra, it must be correct and fully ready for substitution of given values. Do not allow $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2\alpha^2\beta^2$ unless recovered
	A1	For substituting the given values to find the correct value for $\alpha^4+\beta^4$
(b) (ii)	M1	For the correct algebra to write $\alpha^4-\beta^4$ in terms of $(\alpha^2+\beta^2), (\alpha-\beta)$ and $(\alpha+\beta)$
	A1	For the correct value of $\alpha^4-\beta^4$
(c)	M1	For adding together $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate β^4 and reach α^4 either as an expression or implied by adding their values together and dividing by 2. If students subtract $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate α^4 and reach β^4, they must then reach α^4 as an expression or implied by subtracting their values, using one of the expressions from part b and dividing by 2.
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT1	M1	For $\alpha^4 = (3+\sqrt{6})^4$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT2	M1	For $\alpha^4 = \beta^4+360\sqrt{6} = (3-\sqrt{6})^4+360\sqrt{6}$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$

Type MS · The difference of the roots

Answer 1 of 12 · Question 11 part b.ii · 2 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part b.ii in this mark scheme extract

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Type MS · The difference of the roots

Answer 2 of 12 · Question 10 part b · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number		
10(a)	$\alpha+\beta=-3 \quad \alpha\beta=-5$	B1
(i)	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \ =19$	M1,A1
(ii)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \ =19^2-50=311$	M1,A1(5)
	OR: $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2, \ =19^2-50=311$	
(b)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=19+10$ OR $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=9-(-20)$	M1
	$\alpha-\beta=\sqrt{29}$ *	A1 cso (2)
(c)	$\alpha^4-\beta^4=(\alpha^2-\beta^2)(\alpha^2+\beta^2)=(\alpha-\beta)(\alpha+\beta)(\alpha^2+\beta^2)$	M1A1A1 (3)
(d)	$\alpha^4-\beta^4=\sqrt{29}\times(-3)\times19=-57\sqrt{29} \ \left(-\sqrt{94221}\right)$	M1A1 (2)
(e)	$2\beta^4=\alpha^4+\beta^4-(\alpha^4-\beta^4)$	M1
	$\beta^4=\frac{1}{2}(311+57\sqrt{29}), =\frac{311}{2}+\frac{57}{2}\sqrt{29}$	A1,A1 (3)
	$p=\frac{311}{2} \quad q=\frac{57}{2}$	[15]
ALT	$\beta^4=\left(\frac{-3-\sqrt{29}}{2}\right)^4$ **and** use a correct binomial expansion M1A1 Correct final answer A1	

(a)B1	Correct sum **and** product of roots, seen explicitly or used (in (a)). Must be clear that sum is negative
(i)M1	Correct algebra, ready for substitution of sum and product
A1	Correct answer, condone use of $\alpha+\beta=3$.
(ii)M1	Correct algebra, ready for substitution
A1	Correct answer, condone use of $\alpha+\beta=3$.
(b)M1	Correct algebra and substitution of their values
A1cso	Correct answer from correct working. Must have seen sum = -3 here if not shown in (a)
(c)	
M1	Factorise to 2 quadratic brackets or 2 linear and one quadratic bracket
A1	Obtain 2 linear and 1 quadratic brackets with 2 of the 3 brackets correct
A1	Third correct bracket Accept $(\alpha^2+\beta^2)$ or $((\alpha+\beta)^2-2\alpha\beta)$
(d)M1	Substitute their values for each of the 3 brackets obtained in (c)
A1	Correct answer as shown or equivalent **exact** value
(e)	
M1	Correct expression for $2\beta^4$ or β^4
A1ft	Substitute their numbers to obtain a numerical expression for β^4 The expression must be exact but need not be simplified **NB** A correct numerical expression for their values implies M1
A1	Correct answer in the required form. p and q need not be shown explicitly.

Type MS · The difference of the roots

Answer 2 of 12 · Question 10 part b · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks

Type MS · The difference of the roots

Answer 3 of 12 · Question 10 part c · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

Number		
10(a)	$\alpha+\beta=-3 \quad \alpha\beta=-5$	B1
(i)	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \ =19$	M1,A1
(ii)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \ =19^2-50=311$	M1,A1(5)
	OR: $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2, \ =19^2-50=311$	
(b)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=19+10$ OR $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=9-(-20)$	M1
	$\alpha-\beta=\sqrt{29}$ *	A1 cso (2)
(c)	$\alpha^4-\beta^4=(\alpha^2-\beta^2)(\alpha^2+\beta^2)=(\alpha-\beta)(\alpha+\beta)(\alpha^2+\beta^2)$	M1A1A1 (3)
(d)	$\alpha^4-\beta^4=\sqrt{29}\times(-3)\times19=-57\sqrt{29} \ \left(-\sqrt{94221}\right)$	M1A1 (2)
(e)	$2\beta^4=\alpha^4+\beta^4-(\alpha^4-\beta^4)$	M1
	$\beta^4=\frac{1}{2}(311+57\sqrt{29}), =\frac{311}{2}+\frac{57}{2}\sqrt{29}$	A1,A1 (3)
	$p=\frac{311}{2} \quad q=\frac{57}{2}$	[15]
ALT	$\beta^4=\left(\frac{-3-\sqrt{29}}{2}\right)^4$ **and** use a correct binomial expansion M1A1 Correct final answer A1	

(a)B1	Correct sum **and** product of roots, seen explicitly or used (in (a)). Must be clear that sum is negative
(i)M1	Correct algebra, ready for substitution of sum and product
A1	Correct answer, condone use of $\alpha+\beta=3$.
(ii)M1	Correct algebra, ready for substitution
A1	Correct answer, condone use of $\alpha+\beta=3$.
(b)M1	Correct algebra and substitution of their values
A1cso	Correct answer from correct working. Must have seen sum = -3 here if not shown in (a)
(c)	
M1	Factorise to 2 quadratic brackets or 2 linear and one quadratic bracket
A1	Obtain 2 linear and 1 quadratic brackets with 2 of the 3 brackets correct
A1	Third correct bracket Accept $(\alpha^2+\beta^2)$ or $((\alpha+\beta)^2-2\alpha\beta)$
(d)M1	Substitute their values for each of the 3 brackets obtained in (c)
A1	Correct answer as shown or equivalent **exact** value
(e)	
M1	Correct expression for $2\beta^4$ or β^4
A1ft	Substitute their numbers to obtain a numerical expression for β^4 The expression must be exact but need not be simplified **NB** A correct numerical expression for their values implies M1
A1	Correct answer in the required form. p and q need not be shown explicitly.

Type MS · The difference of the roots

Answer 3 of 12 · Question 10 part c · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Scheme	Marks

Type MS · The difference of the roots

Answer 4 of 12 · Question 10 part d · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Number		
10(a)	$\alpha+\beta=-3 \quad \alpha\beta=-5$	B1
(i)	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \ =19$	M1,A1
(ii)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \ =19^2-50=311$	M1,A1(5)
	OR: $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2, \ =19^2-50=311$	
(b)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=19+10$ OR $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=9-(-20)$	M1
	$\alpha-\beta=\sqrt{29}$ *	A1 cso (2)
(c)	$\alpha^4-\beta^4=(\alpha^2-\beta^2)(\alpha^2+\beta^2)=(\alpha-\beta)(\alpha+\beta)(\alpha^2+\beta^2)$	M1A1A1 (3)
(d)	$\alpha^4-\beta^4=\sqrt{29}\times(-3)\times19=-57\sqrt{29} \ \left(-\sqrt{94221}\right)$	M1A1 (2)
(e)	$2\beta^4=\alpha^4+\beta^4-(\alpha^4-\beta^4)$	M1
	$\beta^4=\frac{1}{2}(311+57\sqrt{29}), =\frac{311}{2}+\frac{57}{2}\sqrt{29}$	A1,A1 (3)
	$p=\frac{311}{2} \quad q=\frac{57}{2}$	[15]
ALT	$\beta^4=\left(\frac{-3-\sqrt{29}}{2}\right)^4$ **and** use a correct binomial expansion M1A1 Correct final answer A1	

(a)B1	Correct sum **and** product of roots, seen explicitly or used (in (a)). Must be clear that sum is negative
(i)M1	Correct algebra, ready for substitution of sum and product
A1	Correct answer, condone use of $\alpha+\beta=3$.
(ii)M1	Correct algebra, ready for substitution
A1	Correct answer, condone use of $\alpha+\beta=3$.
(b)M1	Correct algebra and substitution of their values
A1cso	Correct answer from correct working. Must have seen sum = -3 here if not shown in (a)
(c)	
M1	Factorise to 2 quadratic brackets or 2 linear and one quadratic bracket
A1	Obtain 2 linear and 1 quadratic brackets with 2 of the 3 brackets correct
A1	Third correct bracket Accept $(\alpha^2+\beta^2)$ or $((\alpha+\beta)^2-2\alpha\beta)$
(d)M1	Substitute their values for each of the 3 brackets obtained in (c)
A1	Correct answer as shown or equivalent **exact** value
(e)	
M1	Correct expression for $2\beta^4$ or β^4
A1ft	Substitute their numbers to obtain a numerical expression for β^4 The expression must be exact but need not be simplified **NB** A correct numerical expression for their values implies M1
A1	Correct answer in the required form. p and q need not be shown explicitly.

Type MS · The difference of the roots

Answer 4 of 12 · Question 10 part d · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question Number	Scheme	Marks

Type MS · The difference of the roots

Answer 5 of 12 · Question 10 part e · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part e in this mark scheme extract

Number		
10(a)	$\alpha+\beta=-3 \quad \alpha\beta=-5$	B1
(i)	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \ =19$	M1,A1
(ii)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \ =19^2-50=311$	M1,A1(5)
	OR: $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2, \ =19^2-50=311$	
(b)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=19+10$ OR $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=9-(-20)$	M1
	$\alpha-\beta=\sqrt{29}$ *	A1 cso (2)
(c)	$\alpha^4-\beta^4=(\alpha^2-\beta^2)(\alpha^2+\beta^2)=(\alpha-\beta)(\alpha+\beta)(\alpha^2+\beta^2)$	M1A1A1 (3)
(d)	$\alpha^4-\beta^4=\sqrt{29}\times(-3)\times19=-57\sqrt{29} \ \left(-\sqrt{94221}\right)$	M1A1 (2)
(e)	$2\beta^4=\alpha^4+\beta^4-(\alpha^4-\beta^4)$	M1
	$\beta^4=\frac{1}{2}(311+57\sqrt{29}), =\frac{311}{2}+\frac{57}{2}\sqrt{29}$	A1,A1 (3)
	$p=\frac{311}{2} \quad q=\frac{57}{2}$	[15]
ALT	$\beta^4=\left(\frac{-3-\sqrt{29}}{2}\right)^4$ **and** use a correct binomial expansion M1A1 Correct final answer A1	

(a)B1	Correct sum **and** product of roots, seen explicitly or used (in (a)). Must be clear that sum is negative
(i)M1	Correct algebra, ready for substitution of sum and product
A1	Correct answer, condone use of $\alpha+\beta=3$.
(ii)M1	Correct algebra, ready for substitution
A1	Correct answer, condone use of $\alpha+\beta=3$.
(b)M1	Correct algebra and substitution of their values
A1cso	Correct answer from correct working. Must have seen sum = -3 here if not shown in (a)
(c)	
M1	Factorise to 2 quadratic brackets or 2 linear and one quadratic bracket
A1	Obtain 2 linear and 1 quadratic brackets with 2 of the 3 brackets correct
A1	Third correct bracket Accept $(\alpha^2+\beta^2)$ or $((\alpha+\beta)^2-2\alpha\beta)$
(d)M1	Substitute their values for each of the 3 brackets obtained in (c)
A1	Correct answer as shown or equivalent **exact** value
(e)	
M1	Correct expression for $2\beta^4$ or β^4
A1ft	Substitute their numbers to obtain a numerical expression for β^4 The expression must be exact but need not be simplified **NB** A correct numerical expression for their values implies M1
A1	Correct answer in the required form. p and q need not be shown explicitly.

Type MS · The difference of the roots

Answer 5 of 12 · Question 10 part e · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part e in this mark scheme extract

Question Number	Scheme	Marks

Type MS · The difference of the roots

Answer 6 of 12 · Question 6 part a · 3 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Number		
6 (a)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2$	M1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-2\alpha\beta-2\alpha\beta$	M1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *	A1 cso (3)
(b)	$\alpha+\beta=7k$ and $\alpha\beta=k^2$	B1
	$\alpha-\beta=\sqrt{49k^2-4k^2}$	M1
	$=\sqrt{45}k=3k\sqrt{5}$ *	A1 cso (3)
(c)	Sum = $\alpha+\beta$ $(=7k)$	B1ft
	Product $(\alpha+1)(\beta-1)=\alpha\beta-(\alpha-\beta)-1\Rightarrow(\alpha+1)(\beta-1)=k^2-3k\sqrt{5}-1$	M1
	So $x^2-7kx+k^2-3k\sqrt{5}-1=0$	M1A1 (4)
		Total 10 marks

Type MS · The difference of the roots

Answer 6 of 12 · Question 6 part a · 3 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For expanding $(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2\Rightarrow(\alpha^2+\beta^2-2\alpha\beta)$ or expanding $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta$ This must be correct for this mark.
	M1	Replaces $\alpha^2+\beta^2$ with $(\alpha+\beta)^2-2\alpha\beta$ in their expansion of $(\alpha-\beta)^2$. And attempts to collect terms $(\alpha-\beta)^2=(\alpha+\beta)^2-2\alpha\beta-2\alpha\beta$
	A1 cso	For the correct given expression **with no errors seen.** $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *
	ALT	
	M1	For expanding $(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2\Rightarrow(\alpha^2+\beta^2-2\alpha\beta)$ or expanding $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta$ This must be correct for this mark.
	M1	For expanding the RHS and equates to their expansion of $(\alpha-\beta)^2$ and attempting to simplify. $\alpha^2+\beta^2-2\alpha\beta=(\alpha^2+\beta^2+2\alpha\beta)-4\alpha\beta$
	A1 cso	Both sides of the equivalence are shown to be equal **with no errors seen.** $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *
(b)	B1	For both $\alpha+\beta=7k$ and $\alpha\beta=k^2$ This may be implied in later work e.g. by use of $49k^2$ and $4k^2$
	M1	For substituting their values for the sum and product into the **given** expression for $(\alpha-\beta)^2$, simplifying and square rooting both sides. $(\alpha-\beta)^2=(7k)^2-4k^2\Rightarrow\alpha-\beta=\sqrt{(7k)^2-4k^2}=\sqrt{45k^2}$ Condone $\pm\sqrt{45k^2}$ for this mark.
	A1 cso	For the correct value of $\alpha-\beta=3k\sqrt{5}$ *
(c)	B1ft	For the sum $(\alpha+1+\beta-1)=\alpha+\beta=7k$ Ft their $\alpha+\beta$
	M1	For the product in terms of k. Correctly multiplying out $(\alpha+1)(\beta-1)$ and substituting in their value $\alpha\beta$ and the correct value $\alpha-\beta=3k\sqrt{5}$ $(\alpha+1)(\beta-1)=\alpha\beta-(\alpha-\beta)-1\Rightarrow(\alpha+1)(\beta-1)=k^2-3k\sqrt{5}-1$
	M1	For correctly forming an equation with their sum and product $x^2-'7k'x+'k^2-3k\sqrt{5}-1'(=0)$ Condone the absence of = 0 for this mark
	A1	For the correct equation $x^2-7kx+k^2-3k\sqrt{5}-1=0$ Allow $p=-7k\quad q=k^2-3k\sqrt{5}-1$

Type MS · The difference of the roots

Answer 6 of 12 · Question 6 part a · 3 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · The difference of the roots

Answer 7 of 12 · Question 6 part b · 3 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

Question number		
6 (a)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2$	M1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-2\alpha\beta-2\alpha\beta$	M1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *	A1 cso (3)
(b)	$\alpha+\beta=7k$ and $\alpha\beta=k^2$	B1
	$\alpha-\beta=\sqrt{49k^2-4k^2}$	M1
	$=\sqrt{45}k=3k\sqrt{5}$ *	A1 cso (3)
(c)	Sum = $\alpha+\beta$ $(=7k)$	B1ft
	Product $(\alpha+1)(\beta-1)=\alpha\beta-(\alpha-\beta)-1\Rightarrow(\alpha+1)(\beta-1)=k^2-3k\sqrt{5}-1$	M1
	So $x^2-7kx+k^2-3k\sqrt{5}-1=0$	M1A1 (4)
		Total 10 marks

Type MS · The difference of the roots

Answer 7 of 12 · Question 6 part b · 3 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For expanding $(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2\Rightarrow(\alpha^2+\beta^2-2\alpha\beta)$ or expanding $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta$ This must be correct for this mark.
	M1	Replaces $\alpha^2+\beta^2$ with $(\alpha+\beta)^2-2\alpha\beta$ in their expansion of $(\alpha-\beta)^2$. And attempts to collect terms $(\alpha-\beta)^2=(\alpha+\beta)^2-2\alpha\beta-2\alpha\beta$
	A1 cso	For the correct given expression **with no errors seen.** $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *
	ALT	
	M1	For expanding $(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2\Rightarrow(\alpha^2+\beta^2-2\alpha\beta)$ or expanding $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta$ This must be correct for this mark.
	M1	For expanding the RHS and equates to their expansion of $(\alpha-\beta)^2$ and attempting to simplify. $\alpha^2+\beta^2-2\alpha\beta=(\alpha^2+\beta^2+2\alpha\beta)-4\alpha\beta$
	A1 cso	Both sides of the equivalence are shown to be equal **with no errors seen.** $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ *
(b)	B1	For both $\alpha+\beta=7k$ and $\alpha\beta=k^2$ This may be implied in later work e.g. by use of $49k^2$ and $4k^2$
	M1	For substituting their values for the sum and product into the **given** expression for $(\alpha-\beta)^2$, simplifying and square rooting both sides. $(\alpha-\beta)^2=(7k)^2-4k^2\Rightarrow\alpha-\beta=\sqrt{(7k)^2-4k^2}=\sqrt{45k^2}$ Condone $\pm\sqrt{45k^2}$ for this mark.
	A1 cso	For the correct value of $\alpha-\beta=3k\sqrt{5}$ *
(c)	B1ft	For the sum $(\alpha+1+\beta-1)=\alpha+\beta=7k$ Ft their $\alpha+\beta$
	M1	For the product in terms of k. Correctly multiplying out $(\alpha+1)(\beta-1)$ and substituting in their value $\alpha\beta$ and the correct value $\alpha-\beta=3k\sqrt{5}$ $(\alpha+1)(\beta-1)=\alpha\beta-(\alpha-\beta)-1\Rightarrow(\alpha+1)(\beta-1)=k^2-3k\sqrt{5}-1$
	M1	For correctly forming an equation with their sum and product $x^2-'7k'x+'k^2-3k\sqrt{5}-1'(=0)$ Condone the absence of = 0 for this mark
	A1	For the correct equation $x^2-7kx+k^2-3k\sqrt{5}-1=0$ Allow $p=-7k\quad q=k^2-3k\sqrt{5}-1$

Type MS · The difference of the roots

Answer 7 of 12 · Question 6 part b · 3 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · The difference of the roots

Answer 8 of 12 · Question 7 part b · 4 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
7 (a)	$\alpha+\beta=-\frac{p}{2}$ and $\alpha\beta=\frac{q}{2}$	B1
	$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=-\frac{37}{14}$	B1
	$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2+\beta^2}{\alpha\beta}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$	M1
	$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}=\frac{\left(-\frac{p}{2}\right)^2-2\left(\frac{q}{2}\right)}{\frac{q}{2}}=-\frac{37}{14}$	A1
	$\frac{\left(-\frac{p}{2}\right)^2-2\left(\frac{q}{2}\right)}{\frac{q}{2}}=\frac{p^2-4q}{2q}=\frac{p^2+4p+12}{2(-p-4)}=-\frac{37}{14}\Rightarrow 7p^2-9p-36=0$	dM1A1
	$[7p^2-9p-36=0 \quad \text{OR} \quad 7q^2+65q+112=0]$	
	$p=\frac{-(-9)\pm\sqrt{(-9)^2-4\times7\times(-36)}}{2\times7}\Rightarrow p=3,\ q=-7$	M1A1A1ft [9]
(b)	$\alpha^2-\beta^2=(\alpha-\beta)(\alpha+\beta)$	M1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$	M1
	$(\alpha-\beta)(\alpha+\beta)=\sqrt{\left('-\frac{3}{2}'\right)^2-4\times\left('-\frac{7}{2}'\right)}\times\left('-\frac{3}{2}'\right)=-\frac{3\sqrt{65}}{4}$	M1A1 [4]
	Total	**13 marks**

Part	Mark	Notes
(a)	B1	For the sum and product of roots of the equation f(x) = 0 in terms of p and q $\alpha+\beta=-\frac{p}{2}$ and $\alpha\beta=\frac{q}{2}$
	B1	For the sum of roots of the equation g(x) = 0 $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=-\frac{37}{14}$
	M1	For the correct algebra to find the sum of roots of g(x) = 0 ready for substitution of values. $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2+\beta^2}{\alpha\beta}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$

Type MS · The difference of the roots

Answer 8 of 12 · Question 7 part b · 4 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

		This can be implied by correct substitution of their values for the sum and product.
	A1	For the correct value of $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$ in terms of p and q [Simplification not required] $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}=\frac{\left(-\frac{p}{2}\right)^2-2\left(\frac{q}{2}\right)}{\frac{q}{2}}=-\frac{37}{14}$
	dM1	For substituting in for either p or for q into the above equation **correctly** to form a 3TQ in just p or q [ft their $-\frac{37}{14}$ and their $-\frac{p}{2}$ and $\frac{q}{2}$]. The algebra must be correct here. $\frac{\left(-\frac{p}{2}\right)^2-2\left(\frac{q}{2}\right)}{\frac{q}{2}}=\frac{p^2-4q}{2q}=\frac{p^2+4p+12}{2(-p-4)}=-\frac{37}{14}\Rightarrow\left[7p^2-9p-36=0\right]$ This mark is dependent on the previous M mark.
	A1	For the correct 3TQ $7p^2-9p-36=0$ OR $7q^2+65q+112=0$
	M1	For attempting to solve their 3TQ by any valid method $(7p+12)(p-3)=0\Rightarrow p=\ldots\ (p=3)$ OR $p=\frac{-(-9)\pm\sqrt{(-9)^2-4\times7\times(-36)}}{2\times7}\Rightarrow p=\ldots,\ldots$ Negative value of p not required
	A1	For the correct value of $p=3$ **or** value of $q=-7$
	A1ft	For the correct value of $p=3$ **and** value of $q=-7$ Ft on their value of p or q Ignore mislabelling of (i) and (ii) or even no labelling of parts at all.
(b)	M1	For factorising $\alpha^2-\beta^2=(\alpha-\beta)(\alpha+\beta)$ correctly
	M1	For the correct algebra on $(\alpha-\beta)^2$ i.e. $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ Look for this expansion which may be embedded in their working.
	M1	For the correct substitution of their sum and product into $(\alpha-\beta)(\alpha+\beta)$ $(\alpha-\beta)(\alpha+\beta)=\sqrt{\left('-\frac{3}{2}'\right)^2-4\times\left('-\frac{7}{2}'\right)}\times\left('-\frac{3}{2}'\right)=\left[\sqrt{\frac{65}{4}}\times\left(-\frac{3}{2}\right)\right]$
	A1	For the correct answer only $\alpha^2-\beta^2=-\frac{3\sqrt{65}}{4}$

Type MS · The difference of the roots

Answer 8 of 12 · Question 7 part b · 4 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Mark

Type MS · The difference of the roots

Answer 9 of 12 · Question 5 part b · 4 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

5 a	$\alpha+\beta=-\frac{b}{a}=\frac{-(6+2p)}{2}=-3-p$	B1
	$\alpha\beta=\frac{c}{a}=\frac{2p}{2}=p$	B1 (2)
b	$(\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta$	M1
	$=(\alpha+\beta)^2-4\alpha\beta$	M1
	$=(-3-p)^2-4p$	M1
	$=9+2p+p^2$ *	A1 cso (4)
c	$(\alpha-\beta)=3$ implies $(\alpha-\beta)^2=9$	B1
	So $9=9+2p+p^2\Rightarrow 0=2p+p^2\Rightarrow 0=p(2+p)$	M1
	$p=0$ or $p=-2$	A1 (3)
		Total 9 marks

Part	Mark	Notes
(a)	**B1**	For the sum $\alpha+\beta=-3-p$ Need not be simplified. E.g., accept $\frac{-(6+2p)}{2}$
	B1	For the product $\alpha\beta=p$ $\left(\text{Accept } \frac{2p}{2}\right)$
(b)	**M1**	For the correct algebra on $(\alpha-\beta)^2$ $(\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta$ Accept un-simplified and terms in any order
	M1	For the **correct** algebra on $(\alpha-\beta)^2$ using $(\alpha+\beta)^2$ $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ Note: $(\alpha+\beta)^2=9+6p+p^2$ which you may see substituted in without seeing the algebra
	M1	For substituting in their sum and product from part (a) $(\alpha-\beta)^2=(-3-p)^2-4p$
	A1 cso	For obtaining the given result $(\alpha-\beta)^2=9+2p+p^2$ Note: condone absence of seeing the LHS. **Note**: **This is a given result.** There must be **no errors or omissions** (including in algebra) in their work for this mark
(c)	**B1**	For $(\alpha-\beta)^2=9$
	M1	For setting the given answer = 9 $9+2p+p^2=9\Rightarrow 2p+p^2=0\Rightarrow p=\ldots$ Two values.
	A1	For $p=0$ or $p=-2$

Type MS · The difference of the roots

Answer 9 of 12 · Question 5 part b · 4 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks
	$\cos 2A = \cos^2 A - \sin^2 A$	

Type MS · The difference of the roots

Answer 10 of 12 · Question 11 part a.i · 4 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part a.i in this mark scheme extract

11 (a)(i)	$\left(\alpha-\beta=2\sqrt{6}\Rightarrow(\alpha-\beta)^2=24\right)\Rightarrow(\alpha+\beta)^2-4\alpha\beta=24$ oe **(1)**	M1
	$\alpha^2+\beta^2=30\Rightarrow(\alpha+\beta)^2-2\alpha\beta=30$ oe **(2)**	M1
	(2) – (1) $6=2\alpha\beta\Rightarrow\alpha\beta=3$ cso	dM1A1* [4]
ALT1	$\left((\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta\Rightarrow\right)2\alpha\beta=\alpha^2+\beta^2-(\alpha-\beta)^2$ oe	M1
	$30-\left(2\sqrt{6}\right)^2=6$	M1
	$2\alpha\beta=6\Rightarrow\alpha\beta=3$	dM1A1 [4]
(ii)	$30=(\alpha+\beta)^2-2\times3\Rightarrow(\alpha+\beta)^2=36\Rightarrow\alpha+\beta=6$ $[\alpha>\beta>0]$ cso	M1A1* [2]
ALT2 (i)	$\alpha-\beta=2\sqrt{6}\rightarrow\alpha=2\sqrt{6}+\beta$	
	$\alpha^2+\beta^2=30\rightarrow\left(2\sqrt{6}+\beta\right)^2+\beta^2=30$	M1
	$2\beta^2+4\sqrt{6}\beta-6=0$ oe eg $\beta^2+2\sqrt{6}\beta-3=0$	M1
	$(\beta=)\dfrac{-2\sqrt{6}\pm\sqrt{\left(2\sqrt{6}\right)^2-4(1)(-3)}}{2}$ oe $\rightarrow\beta\left(=3-\sqrt{6}\right)$	dM1
	$\alpha=3+\sqrt{6}$	
	$\alpha\beta=\left(3+\sqrt{6}\right)\left(3-\sqrt{6}\right)=9+6\sqrt{3}-6\sqrt{3}-6=3$	A1* [4]
(ii)	$\alpha+\beta=3+\sqrt{6}+3-\sqrt{6}=6$	M1A1 [2]
(b)(i)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2(\alpha\beta)^2=30^2-2\times3^2=882$	M1A1 [2]
(b) ii)	$\alpha^4-\beta^4=\left(\alpha^2+\beta^2\right)\left(\alpha^2-\beta^2\right)=\left(\alpha^2+\beta^2\right)(\alpha-\beta)(\alpha+\beta)$	M1
	$\alpha^4-\beta^4=30\times6\times2\sqrt{6}=360\sqrt{6}$	A1 [2]
(c)	$\left(\alpha^4+\beta^4\right)+\left(\alpha^4-\beta^4\right)=2\alpha^4=882+360\sqrt{6}\Rightarrow\alpha^4=441+180\sqrt{6}$	M1A1 [2]
ALT1	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\alpha=3+\sqrt{6}$	
	$\alpha^4=\left(3+\sqrt{6}\right)^4$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
ALT2	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\beta=3-\sqrt{6}$	
	$\alpha^4=\beta^4+360\sqrt{6}=\left(3-\sqrt{6}\right)^4+360\sqrt{6}$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
		Total 12 marks

Type MS · The difference of the roots

Answer 10 of 12 · Question 11 part a.i · 4 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part a.i in this mark scheme extract

Part	Mark	Notes
(a) (i)	M1	For forming an equation of the form $(\alpha+\beta)^2 \pm p\alpha\beta = 24$ oe or $(\alpha+\beta)^2 \pm t\alpha\beta = 30$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correctly solving their simultaneous equations in $\alpha+\beta$ and $\alpha\beta$ to find a value for $\alpha\beta$, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
ALT1	M1	For forming an equation of the form $q\alpha\beta = \alpha^2+\beta^2-(\alpha-\beta)^2$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correct substitution into their equation, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
(a) (ii)	M1	Correctly uses either of their equations (must be of the required form) to substitute the given value of $\alpha\beta$ and obtains a value for $\alpha+\beta$
	A1*cso	For the correct value of $\alpha+\beta$
ALT2 (i)	M1	For an attempt to eliminate α or β and arrive at an unsimplified quadratic equation in one variable. Allow one error
	M1	For the correct quadratic equation
	dM1	For a **fully correct method** to solve their quadratic equation in α or β, dependent on the 1st method mark
	A1*cso	For correctly finding α and β and showing the minimum steps shown to find $\alpha\beta$
	M1	For finding $\alpha+\beta$ with their values
	A1*cso	For finding $\alpha+\beta$ with the correct values, minimum steps as shown.
(b) (i)	M1	For the correct algebra $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2(\alpha\beta)^2$, if they use more complex algebra, it must be correct and fully ready for substitution of given values. Do not allow $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2\alpha^2\beta^2$ unless recovered
	A1	For substituting the given values to find the correct value for $\alpha^4+\beta^4$
(b) (ii)	M1	For the correct algebra to write $\alpha^4-\beta^4$ in terms of $(\alpha^2+\beta^2), (\alpha-\beta)$ and $(\alpha+\beta)$
	A1	For the correct value of $\alpha^4-\beta^4$
(c)	M1	For adding together $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate β^4 and reach α^4 either as an expression or implied by adding their values together and dividing by 2. If students subtract $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate α^4 and reach β^4, they must then reach α^4 as an expression or implied by subtracting their values, using one of the expressions from part b and dividing by 2.
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT1	M1	For $\alpha^4 = (3+\sqrt{6})^4$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT2	M1	For $\alpha^4 = \beta^4+360\sqrt{6} = (3-\sqrt{6})^4+360\sqrt{6}$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$

Type MS · The difference of the roots

Answer 10 of 12 · Question 11 part a.i · 4 marks

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Type MS · The difference of the roots

Answer 11 of 12 · Question 11 part a.ii · 2 marks

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11 (a)(i)	$\left(\alpha-\beta=2\sqrt{6}\Rightarrow(\alpha-\beta)^2=24\right)\Rightarrow(\alpha+\beta)^2-4\alpha\beta=24$ oe **(1)**	M1
	$\alpha^2+\beta^2=30\Rightarrow(\alpha+\beta)^2-2\alpha\beta=30$ oe **(2)**	M1
	(2) – (1) $6=2\alpha\beta\Rightarrow\alpha\beta=3$ cso	dM1A1* [4]
ALT1	$\left((\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta\Rightarrow\right)2\alpha\beta=\alpha^2+\beta^2-(\alpha-\beta)^2$ oe	M1
	$30-\left(2\sqrt{6}\right)^2=6$	M1
	$2\alpha\beta=6\Rightarrow\alpha\beta=3$	dM1A1 [4]
(ii)	$30=(\alpha+\beta)^2-2\times3\Rightarrow(\alpha+\beta)^2=36\Rightarrow\alpha+\beta=6$ $[\alpha>\beta>0]$ cso	M1A1* [2]
ALT2 (i)	$\alpha-\beta=2\sqrt{6}\rightarrow\alpha=2\sqrt{6}+\beta$	
	$\alpha^2+\beta^2=30\rightarrow\left(2\sqrt{6}+\beta\right)^2+\beta^2=30$	M1
	$2\beta^2+4\sqrt{6}\beta-6=0$ oe eg $\beta^2+2\sqrt{6}\beta-3=0$	M1
	$(\beta=)\dfrac{-2\sqrt{6}\pm\sqrt{\left(2\sqrt{6}\right)^2-4(1)(-3)}}{2}$ oe $\rightarrow\beta\left(=3-\sqrt{6}\right)$	dM1
	$\alpha=3+\sqrt{6}$	
	$\alpha\beta=\left(3+\sqrt{6}\right)\left(3-\sqrt{6}\right)=9+6\sqrt{3}-6\sqrt{3}-6=3$	A1* [4]
(ii)	$\alpha+\beta=3+\sqrt{6}+3-\sqrt{6}=6$	M1A1 [2]
(b)(i)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2(\alpha\beta)^2=30^2-2\times3^2=882$	M1A1 [2]
(b) ii)	$\alpha^4-\beta^4=\left(\alpha^2+\beta^2\right)\left(\alpha^2-\beta^2\right)=\left(\alpha^2+\beta^2\right)(\alpha-\beta)(\alpha+\beta)$	M1
	$\alpha^4-\beta^4=30\times6\times2\sqrt{6}=360\sqrt{6}$	A1 [2]
(c)	$\left(\alpha^4+\beta^4\right)+\left(\alpha^4-\beta^4\right)=2\alpha^4=882+360\sqrt{6}\Rightarrow\alpha^4=441+180\sqrt{6}$	M1A1 [2]
ALT1	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\alpha=3+\sqrt{6}$	
	$\alpha^4=\left(3+\sqrt{6}\right)^4$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
ALT2	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\beta=3-\sqrt{6}$	
	$\alpha^4=\beta^4+360\sqrt{6}=\left(3-\sqrt{6}\right)^4+360\sqrt{6}$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
		Total 12 marks

Type MS · The difference of the roots

Answer 11 of 12 · Question 11 part a.ii · 2 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part a.ii in this mark scheme extract

Part	Mark	Notes
(a) (i)	M1	For forming an equation of the form $(\alpha+\beta)^2 \pm p\alpha\beta = 24$ oe or $(\alpha+\beta)^2 \pm t\alpha\beta = 30$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correctly solving their simultaneous equations in $\alpha+\beta$ and $\alpha\beta$ to find a value for $\alpha\beta$, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
ALT1	M1	For forming an equation of the form $q\alpha\beta = \alpha^2+\beta^2-(\alpha-\beta)^2$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correct substitution into their equation, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
(a) (ii)	M1	Correctly uses either of their equations (must be of the required form) to substitute the given value of $\alpha\beta$ and obtains a value for $\alpha+\beta$
	A1*cso	For the correct value of $\alpha+\beta$
ALT2 (i)	M1	For an attempt to eliminate α or β and arrive at an unsimplified quadratic equation in one variable. Allow one error
	M1	For the correct quadratic equation
	dM1	For a **fully correct method** to solve their quadratic equation in α or β, dependent on the 1st method mark
	A1*cso	For correctly finding α and β and showing the minimum steps shown to find $\alpha\beta$
	M1	For finding $\alpha+\beta$ with their values
	A1*cso	For finding $\alpha+\beta$ with the correct values, minimum steps as shown.
(b) (i)	M1	For the correct algebra $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2(\alpha\beta)^2$, if they use more complex algebra, it must be correct and fully ready for substitution of given values. Do not allow $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2\alpha^2\beta^2$ unless recovered
	A1	For substituting the given values to find the correct value for $\alpha^4+\beta^4$
(b) (ii)	M1	For the correct algebra to write $\alpha^4-\beta^4$ in terms of $(\alpha^2+\beta^2), (\alpha-\beta)$ and $(\alpha+\beta)$
	A1	For the correct value of $\alpha^4-\beta^4$
(c)	M1	For adding together $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate β^4 and reach α^4 either as an expression or implied by adding their values together and dividing by 2. If students subtract $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate α^4 and reach β^4, they must then reach α^4 as an expression or implied by subtracting their values, using one of the expressions from part b and dividing by 2.
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT1	M1	For $\alpha^4 = (3+\sqrt{6})^4$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT2	M1	For $\alpha^4 = \beta^4+360\sqrt{6} = (3-\sqrt{6})^4+360\sqrt{6}$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$

Type MS · The difference of the roots

Answer 11 of 12 · Question 11 part a.ii · 2 marks

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Answer 12 of 12 · Question 11 part c · 2 marks

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11 (a)(i)	$\left(\alpha-\beta=2\sqrt{6}\Rightarrow(\alpha-\beta)^2=24\right)\Rightarrow(\alpha+\beta)^2-4\alpha\beta=24$ oe **(1)**	M1
	$\alpha^2+\beta^2=30\Rightarrow(\alpha+\beta)^2-2\alpha\beta=30$ oe **(2)**	M1
	(2) – (1) $6=2\alpha\beta\Rightarrow\alpha\beta=3$ cso	dM1A1* [4]
ALT1	$\left((\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta\Rightarrow\right)2\alpha\beta=\alpha^2+\beta^2-(\alpha-\beta)^2$ oe	M1
	$30-\left(2\sqrt{6}\right)^2=6$	M1
	$2\alpha\beta=6\Rightarrow\alpha\beta=3$	dM1A1 [4]
(ii)	$30=(\alpha+\beta)^2-2\times3\Rightarrow(\alpha+\beta)^2=36\Rightarrow\alpha+\beta=6$ $[\alpha>\beta>0]$ cso	M1A1* [2]
ALT2 (i)	$\alpha-\beta=2\sqrt{6}\rightarrow\alpha=2\sqrt{6}+\beta$	
	$\alpha^2+\beta^2=30\rightarrow\left(2\sqrt{6}+\beta\right)^2+\beta^2=30$	M1
	$2\beta^2+4\sqrt{6}\beta-6=0$ oe eg $\beta^2+2\sqrt{6}\beta-3=0$	M1
	$(\beta=)\dfrac{-2\sqrt{6}\pm\sqrt{\left(2\sqrt{6}\right)^2-4(1)(-3)}}{2}$ oe $\rightarrow\beta\left(=3-\sqrt{6}\right)$	dM1
	$\alpha=3+\sqrt{6}$	
	$\alpha\beta=\left(3+\sqrt{6}\right)\left(3-\sqrt{6}\right)=9+6\sqrt{3}-6\sqrt{3}-6=3$	A1* [4]
(ii)	$\alpha+\beta=3+\sqrt{6}+3-\sqrt{6}=6$	M1A1 [2]
(b)(i)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2(\alpha\beta)^2=30^2-2\times3^2=882$	M1A1 [2]
(b) ii)	$\alpha^4-\beta^4=\left(\alpha^2+\beta^2\right)\left(\alpha^2-\beta^2\right)=\left(\alpha^2+\beta^2\right)(\alpha-\beta)(\alpha+\beta)$	M1
	$\alpha^4-\beta^4=30\times6\times2\sqrt{6}=360\sqrt{6}$	A1 [2]
(c)	$\left(\alpha^4+\beta^4\right)+\left(\alpha^4-\beta^4\right)=2\alpha^4=882+360\sqrt{6}\Rightarrow\alpha^4=441+180\sqrt{6}$	M1A1 [2]
ALT1	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\alpha=3+\sqrt{6}$	
	$\alpha^4=\left(3+\sqrt{6}\right)^4$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
ALT2	$\alpha-\beta=2\sqrt{6},\alpha+\beta=6\Rightarrow\beta=3-\sqrt{6}$	
	$\alpha^4=\beta^4+360\sqrt{6}=\left(3-\sqrt{6}\right)^4+360\sqrt{6}$	M1
	$\alpha^4=441+180\sqrt{6}$	A1 [2]
		Total 12 marks

Type MS · The difference of the roots

Answer 12 of 12 · Question 11 part c · 2 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a) (i)	M1	For forming an equation of the form $(\alpha+\beta)^2 \pm p\alpha\beta = 24$ oe or $(\alpha+\beta)^2 \pm t\alpha\beta = 30$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correctly solving their simultaneous equations in $\alpha+\beta$ and $\alpha\beta$ to find a value for $\alpha\beta$, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
ALT1	M1	For forming an equation of the form $q\alpha\beta = \alpha^2+\beta^2-(\alpha-\beta)^2$ oe
	M1	For this equation being fully correct. Both marks may be implied by early substitution of values.
	dM1	For correct substitution into their equation, dependent on 1st method mark.
	A1*cso	For the correct value of $\alpha\beta$
(a) (ii)	M1	Correctly uses either of their equations (must be of the required form) to substitute the given value of $\alpha\beta$ and obtains a value for $\alpha+\beta$
	A1*cso	For the correct value of $\alpha+\beta$
ALT2 (i)	M1	For an attempt to eliminate α or β and arrive at an unsimplified quadratic equation in one variable. Allow one error
	M1	For the correct quadratic equation
	dM1	For a **fully correct method** to solve their quadratic equation in α or β, dependent on the 1st method mark
	A1*cso	For correctly finding α and β and showing the minimum steps shown to find $\alpha\beta$
	M1	For finding $\alpha+\beta$ with their values
	A1*cso	For finding $\alpha+\beta$ with the correct values, minimum steps as shown.
(b) (i)	M1	For the correct algebra $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2(\alpha\beta)^2$, if they use more complex algebra, it must be correct and fully ready for substitution of given values. Do not allow $\alpha^4+\beta^4 = (\alpha^2+\beta^2)^2 - 2\alpha^2\beta^2$ unless recovered
	A1	For substituting the given values to find the correct value for $\alpha^4+\beta^4$
(b) (ii)	M1	For the correct algebra to write $\alpha^4-\beta^4$ in terms of $(\alpha^2+\beta^2), (\alpha-\beta)$ and $(\alpha+\beta)$
	A1	For the correct value of $\alpha^4-\beta^4$
(c)	M1	For adding together $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate β^4 and reach α^4 either as an expression or implied by adding their values together and dividing by 2. If students subtract $\alpha^4+\beta^4$ and $\alpha^4-\beta^4$ to eliminate α^4 and reach β^4, they must then reach α^4 as an expression or implied by subtracting their values, using one of the expressions from part b and dividing by 2.
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT1	M1	For $\alpha^4 = (3+\sqrt{6})^4$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$
ALT2	M1	For $\alpha^4 = \beta^4+360\sqrt{6} = (3-\sqrt{6})^4+360\sqrt{6}$
	A1	For the correct value of $\alpha^4 = 441+180\sqrt{6}$

Type MS · The difference of the roots

Answer 12 of 12 · Question 11 part c · 2 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

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